Subsequent musculoskeletal injury after concussion in National Football League players

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ABSTRACT

Objective To assess whether National Football League (NFL) players diagnosed with a concussion have an increased risk of injury after return to football.

Methods A retrospective cohort study analysed the hazard of subsequent time-loss lower extremity (LEX) or any musculoskeletal injury among NFL players diagnosed with a concussion in 2015–2021 preseason or regular season games compared with: (1) all non-concussed players participating in the same game and (2) players with time-loss upper extremity injury. Cox proportional hazards models were adjusted for number of injuries and concussions in the prior year, player tenure and roster position. Additional models accounted for time lost from participation after concussion.

Results There was no statistical difference in the hazards of LEX injury or any musculoskeletal injury among concussed players compared with non-concussed players, though concussed players had a slightly elevated hazard of injury (LEX injury: HR=1.12, 95% CI 0.90 to 1.41; any musculoskeletal injury: HR=1.08, 95% CI: 0.89 to 1.31). When comparing to players with upper extremity injuries, the hazard of injury for concussed players was not statistically different, though HRs suggested a lower injury risk among concussed players (LEX injury: HR=0.78, 95% CI: 0.60 to 1.02; any musculoskeletal injury: HR=0.82, 95% CI: 0.65 to 1.04).

Conclusion We found no statistical difference in the risk of subsequent injury among NFL players returning from concussion compared with non-concussed players in the same game or players returning from upper extremity injury. These results suggest deconditioning or other factors associated with lost participation time may explain subsequent injury risk in concussed players observed in some settings after return to play.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ There are contrasting reports regarding the risk of subsequent musculoskeletal injury following concussion, with few studies having robust data on potential confounders (eg, prior injury history, time at risk), detailed follow-up, or a comparative group of athletes without concussion but with time loss from sport.

WHAT THIS STUDY ADDS

⇒ This study identified time lost due to injury, and not necessarily concussion, as a potential driver of risk for subsequent musculoskeletal injury after return to participation among professional American football players.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Clinicians should consider the increased risk of subsequent injury from deconditioning due to time away from sport, regardless of the cause of the time loss. These findings suggest that concussed athletes as well as those returning from an upper extremity injury may benefit from more robust conditioning prior to reintroduction to sport; the impact of such an approach on all injured athletes should be further investigated.

INTRODUCTION

Concussion can be associated with transient alterations in neuromuscular control, cognitive processing speed, and vestibular and dynamic proprioceptive function.[1 2] This has led to analyses of risk of subsequent injury after return to play following concussion recovery. Examining the association between concussion and future injury risk is complex. While some studies have reported an increased risk of subsequent injury post-concussion, others have failed to support this association.[3–21] Despite mixed findings, three published systematic reviews of the relationship between concussion and lower extremity (LEX) injuries have all concluded that odds or risk of sustaining a LEX injury is higher after a concussion, with effect sizes from 1.60 to 2.49 (ORs and incidence rate ratios).[22–24]

Most published studies do not use data sources generated from robust reporting from medical staff on injuries that occur across the entire year, rather than during the competitive season only, which limits the ability to account for one of the main predictors of injury in sport: previous injury.[25–27] Additionally, other factors such as player age, experience and timing during the competitive season are not taken into account. Many studies also do not accurately account for time at risk after concussion and instead include the time the player misses due to the concussion in the time at risk for a subsequent injury. Selection of the comparison (control) group is also important, as inclusion of only a non-concussed, primarily uninjured comparison group may result in bias since concussed players are removed from sports participation for a period of time while non-concussed, uninjured controls are not.[28 29]

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This study aimed to assess whether National Football League (NFL) players who were diagnosed with concussion had increased risk of subsequent time-loss injury (LEX-specific and overall injury risk) using internal, medically reported data and robust methodology, including detailed injury information, athlete-specific information and multiple comparison groups.

METHODS

Data sources and study population

NFL player injury data were obtained from Club medical staff and entered into the League-wide electronic health record (EHR). Data entry for all sports-related injuries that receive medical treatment is mandatory for all 32 Clubs, using a customised coding system of diagnosis terms specifically tailored to NFL player injuries and adhering to standardised collection guidelines. Data quality reviews, including checks for completeness and consistency, are conducted on a weekly basis to ensure accurate and timely reporting.[30] In addition to reporting injury diagnosis, athletic trainers record additional details of the injury, including setting, timing during the season, player tenure, player position and activity. Data reporting for injuries that result in the player being restricted or limited from their normal level of participation in any activity (ie, 'time-loss injuries') also includes dates of removal, participation status, and the date of return to partial and full football activities. Exposure information on the game and play-level (type of play and total number of plays) for individual athletes is collected from the NFL Game Statistics and Information System for all NFL games and linked to the NFL EHR.

The Concussion in Sport Group's definition of concussion was used throughout the study period.[31 32] Consistent with the NFL Head, Neck and Spine Committee's Concussion Diagnosis and Management Protocol, all players diagnosed with concussion follow a stepwise progression back to full participation which requires final clearance by the Club Team Physician and confirmation by an independent neurological consultant.[33 34] To assess subsequent injury risk after concussion, this study retrospectively compared the hazard of subsequent injury among concussed players to two different comparator groups: (1) all other non-concussed players who participated in the same game (Comparison Group 1) and (2) players who sustained a time-loss upper extremity injury (Comparison Group 2). Concussed players included all players who were diagnosed with a concussion during a 2015–2021 preseason or regular season game, including during pre-game activities. Comparison Group 1 included all players from the same Club as the concussed player who participated in at least one play in the game in which a concussion was sustained, regardless of whether they sustained an injury other than a concussion during that game. This group represented individuals with a similar follow-up time, schedule and Club environment as the concussed group. Comparison Group 2 included players who sustained at least one in-game time-loss upper extremity injury, defined as a sprain, strain or fracture to the arm, clavicle, elbow, fingers, forearm, hand, shoulder or wrist. This group was selected with the hypothesis that these players lost time after their upper extremity injury, similar to the concussed group, but in most cases the upper extremity injury would not be a risk factor for the study outcome of LEX injury. To minimise potential bias, players (in any group) were excluded if they had sustained a concussion within the same season prior to study entry (eg, during practice), did not return to football participation after the index injury or had a missing return from injury date in the EHR (most often due to the player leaving the Club with which the index injury occurred before injury resolution). Players in their rookie season were excluded due to limited prior injury data, which is a strong risk factor for the subsequent injury outcomes of this study.

Outcome measurements

The hazard of subsequent injury was measured for two different musculoskeletal injury outcomes: (1) time-loss LEX injury, defined as a sprain, strain or fracture to the ankle, foot, heel, hip, knee, leg, thigh or toe that resulted in the player being restricted or limited from their normal level of participation and (2) time-loss injury, defined as a sprain, strain or fracture that resulted in being restricted or limited from the player's normal level of participation. These outcome injuries could occur in any football-related setting (eg, practice, game, conditioning session). Outcome injuries were restricted to musculoskeletal injuries (sprain, strain or fracture) because these were considered the injuries on which post-concussion neuromuscular control deficits would have an impact.

Study follow-up

Players entered the study (index date) on the game date of their injury (Concussed Group, Comparison Group 2 and injured players from Comparison Group 1; figure 1, Players 1–4) or the game date in which another player from the same Club sustained a concussion (uninjured players from Comparison Group 1; figure 1, Players 5–7). Follow-up time for injured players began on the day they returned to full participation; the days the player missed due to the concussion or other injury were excluded from follow-up time but accounted for in adjusted models to account for time they were not participating. The hazard of subsequent injury was measured through the remainder of the preseason and regular season (referred to as full-season follow-up), as well as through the 30-day and 60-day periods following the index date.

Study follow-up time was calculated as the number of plays in which the player participated in games during the follow-up period (player-plays), which accounts for exposure to injury risk at a more granular level than follow-up time in days or number of games played. For analyses with any injury as the outcome, study follow-up time ended at the time of injury; if a player was not injured, follow-up was stopped (censored) at the end of the time-period of interest (full-season, 60-day or 30-day) or removal from the team's roster, whichever occurred first. Injured players who were removed from the roster, sustained a subsequent injury prior to return to play or returned to play after the end of the regular season (the end of study follow-up) were excluded from analyses.

For analyses examining time to LEX injury, study follow-up time was calculated the same as described above, except follow-up was stopped for players who sustained an injury other than the outcome injury. Follow-up was not stopped for non-time-loss upper extremity injuries, as occurrence of this injury likely would not influence subsequent risk of LEX injury.

Statistical methods

Descriptive statistics were calculated for all variables of interest including means, medians, SD and IQRs. HRs and 95% CIs were calculated using Cox Proportional Hazards models for the two comparator groups (Comparison Group 1 and Group 2) across both outcomes (time-loss LEX musculoskeletal injury and any time-loss musculoskeletal injury), resulting in four primary statistical models. Sensitivity analyses examined the 30-day and 60-day follow-up period to determine whether there was an

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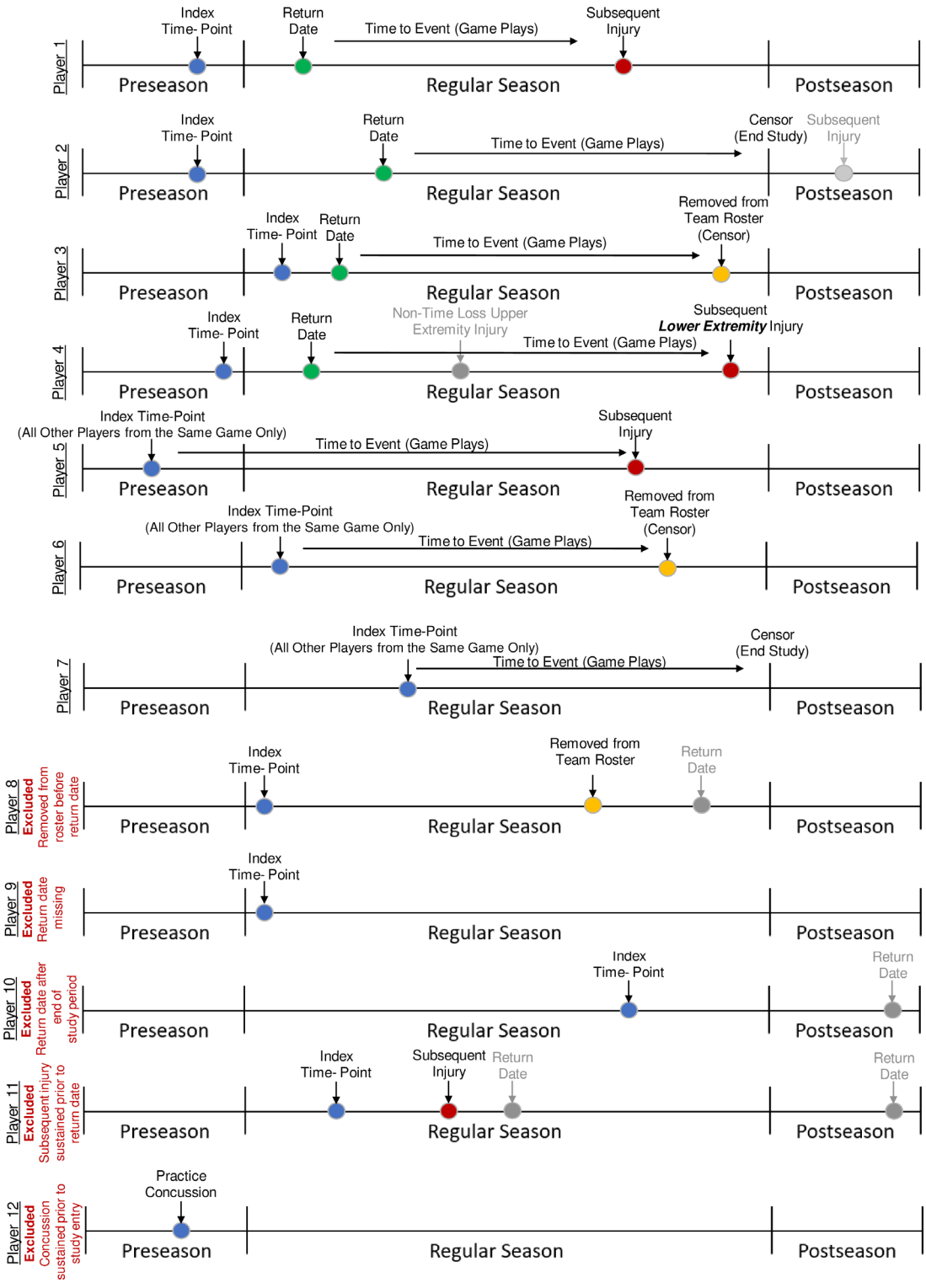

Figure 1 Inclusion and exclusion criteria and calculation of time to event for different cohorts and timing scenarios.[a] [a]Secondary analyses that censored at 30 days and 60 days after the index time-point were also conducted.

impact of concussion on injury risk within the time immediately following concussion, when one would expect a larger deficit in neuromuscular control. A 95% CI exclusive of 1.0 was considered statistically significant.

All models adjusted for number of football-related injuries (all or LEX only in models with LEX injury as the outcome) sustained during the 365 days prior to the index time point (regardless of the setting during which they occurred), the number of football-related concussions during the 365 days prior to the index time point (regardless of the setting during which they occurred), player tenure (years of experience) in the NFL and roster position. Analyses that included number of days missed due to injury from the index date as a potential mediator were also conducted to assess the impact of time missed from participation on the outcome. To account for multiple observations among the same player, we added a frailty term to the models.

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Table 1 Player characteristics by exposure/comparator group, National Football League, preseason and regular season, 2015–2021

Player characteristic	Concussed players (n=766*)	Comparison group 1: all other non-concussed players from the same game (n=51 572*)	Comparison group 2: players with time-loss upper extremity injury† (n=789*)
Age (years); mean (SD)	27.0 (2.9)	27.2 (3.1)	27.2 (3.0)
Weight (lbs); mean (SD)	235.2 (44.8)	246.5 (46.3)	250.0 (47.4)
Tenure (years); mean (SD)	4.0 (2.7)	4.2 (3.0)	4.2 (2.9)
Number of prior injuries‡; mean (SD)	3.8 (2.8)	3.0 (2.7)	3.8 (2.9)
Roster position§; n (%)			
Quarterback	36 (4.7)	1644 (3.2)	49 (6.2)
Running back	65 (8.5)	3933 (7.6)	58 (7.4)
Wide receiver	106 (13.8)	5554 (10.8)	96 (12.2)
Defensive secondary	222 (29.0)	9614 (18.6)	136 (17.2)
Linebacker	85 (11.1)	7166 (13.9)	118 (15.0)
Offensive line	114 (14.9)	8598 (16.7)	138 (17.5)
Defensive line	61 (8.0)	7970 (15.5)	141 (17.9)
Tight end	69 (9.0)	3363 (6.5)	47 (6.0)
Punter/kicker/long snapper	8 (1.0)	3730 (7.2)	6 (0.8)

*A single player could have multiple follow-up periods included in analyses.
†Upper extremity injuries include injuries to the arm, clavicle, elbow, fingers, forearm, hand, shoulder or wrist that resulted in a sprain, strain or fracture.
‡Includes the number of time-loss and non-time-loss injuries sustained by the player within 365 days prior to the index time-point.
§Represents roster position, noting that position at time of injury may be different than roster position, for example, during special teams plays.

Patient and public involvement

Patients and/or the public were not directly involved in the design, conduct or reporting of this research; however, the NFL Players Association approved this work through the NFL Player Scientific and Medical Research Protocol and the Players Association provided feedback on the findings of the work.[35]

Equity, diversity and inclusion statement

All players in the NFL were eligible for inclusion in analyses. All players are men and are racially and ethnically diverse. The research team comprises a diverse group of clinicians and epidemiologists (60% women). Authors were invited based on their expertise, as demonstrated by previous research and clinical experiences with musculoskeletal injury and concussions.

RESULTS

Study population and player characteristics

The dataset included 766 player-observations (defined as the number of non-distinct players included in the analysis) among 641 unique players with a concussion, 51 572 player-observations among 4878 unique players in Comparison Group 1 (non-concussed players from the same game) and 789 player-observations among 653 unique players from Comparison Group 2 (time-loss upper extremity injury). Player age, weight, NFL tenure and number of prior injuries were similar among groups; roster position varied slightly by group (table 1).

Comparison with all non-concussed players from the same game

Concussed players had a slightly smaller proportion of players sustain a subsequent LEX injury across all follow-up time points (table 2). When accounting for potential confounders and taking into account time at risk, there was not a statistically significant difference in the hazard of sustaining a subsequent LEX injury within the same season among concussed players compared with non-concussed players (Comparison Group 1), although the hazard was 9% higher (HR=1.09, 95% CI: 0.88 to 1.35; table 3). Findings were consistent at the 60-day follow-up time

Table 2 Players who sustained a sprain, strain or fracture requiring lost playing time during follow-up by group, National Football League preseason and regular season, 2015–2021

	End of season follow-up		60-day follow-up		30-day follow-up		
	Injury sustained	Injury not sustained	Injury sustained	Injury not sustained	Injury sustained	Injury not sustained	Total
Outcome: subsequent time-loss* lower extremity sprain, strain or fracture							
Concussion	160 (21%)	606 (79%)	136 (18%)	630 (82%)	91 (12%)	675 (88%)	766
Comparison Group 1: other non-concussed players from the same game	11 124 (22%)	40 448 (78%)	9599 (19%)	41 973 (81%)	6797 (13%)	44 775 (87%)	51 572
Comparison Group 2: time-loss upper extremity injury†	207 (26%)	582 (74%)	177 (22%)	612 (78%)	105 (13%)	684 (87%)	789
Outcome: subsequent time-loss* sprain, strain or fracture							
Concussion	268 (35%)	498 (65%)	239 (31%)	527 (69%)	161 (21%)	605 (79%)	766
Comparison Group 1: other non-concussed players from the same game	18 197 (35%)	33 375 (65%)	16 074 (31%)	35 498 (69%)	11 766 (23%)	39 806 (77%)	51 572
Comparison Group 2: time-loss upper extremity injury†	311 (39%)	478 (61%)	267 (34%)	522 (66%)	165 (21%)	624 (79%)	789

*A 'time-loss injury' was defined as an injury that resulted in the player being restricted or limited from their normal level of participation.
†Upper extremity injury includes time-loss injuries to the arm, clavicle, elbow, fingers, forearm, hand, shoulder or wrist that resulted in a sprain, strain or fracture.

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Table 3 Comparison Group 1: hazard of subsequent injury following concussion compared with all other non-concussed players who played in the same game, by outcome; National Football League, preseason and regular season, 2015–2021

Follow-up end	Inclusion of days missed following index date	Adjusted HR*
Outcome: subsequent time-loss lower extremity sprain, strain or fracture		
End of season	No	1.09 (95% CI: 0.88 to 1.35)
	Yes	1.12 (95% CI: 0.90 to 1.41)
60 days	No	1.16 (95% CI: 0.93 to 1.44)
	Yes	1.15 (95% CI: 0.91 to 1.46)
30 days	No	**1.39 (95% CI: 1.07 to 1.82)**
	Yes	1.22 (95% CI: 0.90 to 1.65)
Outcome: subsequent time-loss sprain, strain or fracture		
End of season	No	1.06 (95% CI: 0.88 to 1.27)
	Yes	1.08 (95% CI: 0.89 to 1.31)
60 days	No	1.13 (95% CI: 0.94 to 1.37)
	Yes	1.10 (95% CI: 0.90 to 1.35)
30 days	No	**1.36 (95% CI: 1.08 to 1.70)**
	Yes	1.13 (95% CI: 0.88 to 1.46)

*Adjusted for number of prior injuries (or number of lower extremity injuries in models with lower extremity injury as the outcome) sustained during the 365 days prior to the index time point, the number of prior concussions during the 365 days prior to the index time point, player tenure in the NFL and roster position. Statistically significant results (p<0.05) appear in the table as bolded text.

Table 4 Comparison Group 2: hazard of subsequent injury following concussion compared with time-loss upper extremity injury, National Football League 2015–2021

Follow-up end	Inclusion of days missed following index date	Adjusted HR*
Outcome: subsequent time-loss lower extremity sprain, strain or fracture		
End of season	No	0.79 (95% CI: 0.60 to 1.02)
	Yes	0.78 (95% CI: 0.60 to 1.02)
60 days	No	0.74 (95% CI: 0.56 to 0.98)
	Yes	**0.74 (95% CI: 0.56 to 0.98)**
30 days	No	0.82 (95% CI: 0.57 to 1.16)
	Yes	0.79 (95% CI: 0.56 to 1.14)
Outcome: subsequent time-loss sprain, strain or fracture		
End of season	No	0.83 (95% CI: 0.66 to 1.04)
	Yes	0.82 (95% CI: 0.65 to 1.04)
60 days	No	0.80 (95% CI: 0.63 to 1.02)
	Yes	0.80 (95% CI: 0.63 to 1.02)
30 days	No	0.79 (95% CI: 0.59 to 1.06)
	Yes	0.78 (95% CI: 0.58 to 1.05)

*Adjusted for number of prior injuries (or number of lower extremity injuries in models with lower extremity injury as the outcome) sustained during the 365 days prior to the index time point, the number of prior concussions during the 365 days prior to the index time point, player tenure in the NFL and roster position. Statistically significant results (p<0.05) appear in the table as bolded text.

period, with similar but slightly larger effect estimate. At the 30-day follow-up, when not accounting for time missed due to injury, the concussed group had a significantly higher hazard of subsequent LEX injury (HR=1.39, 95% CI: 1.07 to 1.82; table 3).

When expanding to all time-loss musculoskeletal injuries, concussed players and Comparison Group 1 had the same proportion of players sustain a subsequent time-loss injury when followed through the full-season and at 60 days of follow-up (table 2). At 30 days of follow-up, concussed players had a smaller proportion of players sustain a subsequent time-loss injury compared with Comparison Group 1 (table 2). The hazard of sustaining a subsequent time-loss injury within the same season was not significantly different compared with Comparison Group 1, though it was 6% higher (HR=1.06, 95% CI: 0.88 to 1.27; table 3). At the 30-day follow-up, when not accounting for time missed due to injury, the concussed group had a significantly higher hazard of subsequent musculoskeletal injury (HR=1.36, 95% CI: 1.08 to 1.70; table 3).

Importantly, concussed players lost time due to the concussion before they returned to participation and started their follow-up period for subsequent injury (mean=10.5 days, SD=7.2; median=10.0 days, IQR: 0–81), whereas non-concussed players in Comparison Group 1 only lost time after their index date if they sustained a non-concussion injury during the same game (mean=0.5 days, SD=34; median=0.0 days, IQR=0–93). For the 30-day and 60-day follow-up periods, accounting for this difference shifted the hazard observed for subsequent time-loss injury closer to the null, and the effect at 30 days was no longer statistically significant (table 3).

Comparison with players with time-loss upper extremity injury

In terms of time lost due to the upper extremity injury before the player returned to participation and started their follow-up period for subsequent injury, players in Comparison Group 2 lost an average of 13.7 days (mean=13.7, SD=15.0; median=7.0, IQR=0.0–121.0), which was similar to the 10.5 day average among concussed players.

The hazard of a subsequent time-loss LEX injury within the same season among concussed players was not significantly different compared with players in Comparison Group 2, although the hazard was 22% lower (HR=0.78, 95% CI: 0.60 to 1.02; table 4). At the 60-day follow-up, there was a statistically significantly lower hazard among concussed players (HR=0.74, 95% CI: 0.56 to 0.98). These results were consistent with a non-statistically significant lower hazard of a subsequent time-loss injury for concussed players when compared with players returning after an upper extremity injury (Comparison Group 2; HR=0.82, 95% CI: 0.65 to 1.04; table 4). Results were consistent among 60-day and 30-day follow-up time periods (table 4).

DISCUSSION

In this study of NFL athletes, risk of subsequent musculoskeletal injury was elevated compared with non-concussed athletes but lower compared with players who sustained an upper extremity injury, though neither association was statistically significant. Comparison Group 2 players, who, similar to the concussed players, lost an average of 10–15 days between their upper extremity injury and their return to participation, had a higher hazard of subsequent injury compared with athletes returning from concussion.

There has been significant interest in the relationship between concussion and subsequent injury risk, with literature focused largely on post-concussive LEX injuries. A number of these studies report an increase in injury after concussion, but these frequently use an uninjured control group, which does not account for days lost due to concussion, nor other unmeasured residual confounding (eg, exposure to contact). Other methodological limitations in prior reports include use of publicly available or incomplete injury data, which does not contain

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confirmed or granular return to participation information, accurate (or any) medical history or necessary player-specific information. Importantly, other studies either do not account for follow-up time when the player is at risk for sustaining a subsequent injury or measure this in days rather than using a metric that incorporates true player exposure such as player-plays.[36 37] Of note, Wilson *et al*[8] used injured comparison groups and did not observe a statistically significant elevated risk of subsequent lower extremity injury among concussed players, similar to our findings.

The current consensus within the literature is that concussion interferes with neurological function (eg, cognition, oculomotor, vestibular and balance), leading athletes to greater risk of subsequent injury because of residual deficits in these domains; however, our data suggest that the observed increased risk could be due to the time missed due to concussion. After accounting for time missed due to injury, we observed a slightly elevated risk of subsequent musculoskeletal injury compared with non-concussed players, but this was not statistically significant. If residual neurocognitive effects of concussion are the driving cause for increased subsequent musculoskeletal injury risk, we would have expected to see an increased risk compared with players with upper extremity injuries, which we did not.[1]

Other factors such as physical deconditioning may explain the elevated hazard of injury after injury; however, we neither defined nor measured deconditioning in this study. The models in the current analysis that included days lost following the concussion showed an attenuated subsequent injury risk among concussed players, compared with models that did not include this factor. Early exercise is encouraged after concussion as well as any upper extremity injury; however, concussion return to participation protocols in the NFL specifically require a league-standardised, rigorous, highly-monitored stepwise progression back to full sport, which provides some level of conditioning and, in the football setting, introduction to contact.[34] Practices for return from upper extremity injuries may also be rigorous, but they are more variable based on the type of injury, player position and the medical practitioner. This may explain the observed slightly lower hazard of subsequent time-loss injury among concussed players compared with those with an upper extremity injury.

It is important to note that this study was among elite, male American football athletes, who may have a different risk of subsequent injury than other cohorts. However, our results showed a smaller magnitude of association than reported in other elite sports including European soccer[4] and rugby union,[3] and our results were not statistically significant. In previous studies of NFL players, most authors used publicly available data to assess subsequent risk injury post-concussion and had varying findings, likely due to differences in methods.[17–19 21] Buckley *et al*[18] reported that 21.4% of concussed players had subsequent musculoskeletal injuries in the remainder of the season compared with 26.4% of control athletes, and there was no observed difference in time to subsequent musculoskeletal injury between groups. Similarly, Jildeh *et al*[19] analysed publicly available data for the 2012–2016 seasons and did not observe a statistically significant difference in the odds of LEX injury during the 90-day period after return to play between concussed and control athletes (OR=0.573, 95% CI: 0.270 to 1.217). However, Baker *et al*[17] evaluated all active NFL players during the 2016–2019 seasons and did find players with a single concussion and multiple concussions to have an increased odds of sustaining a LEX injury compared with non-concussed players (OR: 2.92, SD: 1.7–4.9 and OR: 2.28, SD: 1.5–3.6, respectively). These studies did not have the robust data derived from the prospectively maintained and audited injury reports and electronic medical record mandated by NFL policy and procedures. Therefore, they were likely unable to accurately capture all subsequent injuries or account for player history of prior concussion or musculoskeletal injury, nor were they able to accurately measure the time lost due to injury. Pietrosimone *et al*[20] surveyed former NFL athletes about their injury history and showed that players with a history of concussions had higher odds of musculoskeletal injury; however, this did not account for when the injuries occurred relative to each other or time at risk.

Clinical implications

Findings of the current study may suggest that time lost due to injury, and not necessarily lingering neurologic dysfunction, may be the driving factor in subsequent injury risk after return to play. This observation was also noted in athletes returning from time lost due to an upper extremity injury in the absence of concussion. Taken together, our findings suggest that concussed athletes may benefit from rigorous, protocolised approach to reintroduction to sport; we recommend further research to determine whether such an approach could have a beneficial impact on all injured athletes.

Limitations

Limitations of the current study include differences in follow-up time across the athletes after return to play, especially for those athletes in Comparison Group 1 who were not injured at the start of follow-up. Athletes were censored at the end of regular season or removal from the team's roster, which resulted in follow-up for some athletes being short of the 30-day, 60-day or full-season follow-up time-periods; however, these censorship points are accounted for by the model and are considered to be uninformative of the outcome of interest. The use of in-game player-plays as a measurement of follow-up is a strength of our study, as it accounts for the player's true exposure to football participation; however, it does not account for exposure across all types of football activities (eg, practices, pre-game). Additionally, injury history prior to the NFL (such as collegiate injury history) was not available; as a result, rookie players were excluded from these analyses and models were adjusted for prior injury only in the 365 days prior to the index time point. This study covers a wide time range, during which concussion detection and management changed; however, sensitivity analyses did not demonstrate a change in results interpretation when restricting to more recent years (online supplemental table). This study was able to demonstrate the impact of using different comparison groups and control for a number of potential confounding factors; however, we could not fully account for intrinsic injury risk or potential for different injury risk based on game schedule (eg, opponent, weather). The applicability of these findings to other sports with a different season schedule, non-elite athletes or women is not clear and could not be assessed with this study.

CONCLUSION

Results demonstrate that, in elite male American football athletes, risk of subsequent musculoskeletal injury was elevated compared with non-concussed athletes but lower compared with players who sustained an upper extremity injury, though neither association was statistically significant. These findings are distinct from some previous studies due to the ability to account for prior injuries as well as time out of sport during recovery and time at risk following return. The results highlight the need for increased attention to reintroduction to sport after any injury, across all sports, as time lost due to injury may be an important factor in risk of subsequent injury.

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Contributors EW, AC, MH and CM had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis, EW as guarantor. All authors contributed to the concept and design of the manuscript, results interpretation drafting of the manuscript and critical revision of the manuscript for important intellectual content. EW, MH and AC completed the acquisition, analysis or interpretation of data as well as the statistical analysis. EW, MH, CM and AC provided administrative, technical or material support. CM, MH and EW provided analytic supervision.

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Competing interests EW, AC, MH and CM are employees of IQVIA, which is in a paid consultancy with the NFL. GSS is a paid consultant to the NFL. MP is a Senior Advisor for the NFL Head Neck & Spine Committee, a member of the NFL General Medical Committee and a paid consultant to MLS. JC is the Vice Chair of the NFL Head Neck & Spine Committee and a paid consultant to the NFL. AS is a full-time employee of the NFL.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

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